

Patient Monitoring System using Hand Gestures

Abstract: This paper presents a novel patient monitoring system that utilizes hand gestures for interaction. The system is designed to monitor vital signs and patient status in real-time, providing a non-invasive and intuitive interface for healthcare professionals. The system architecture includes a camera for gesture recognition, a processing unit for data analysis, and a display for real-time monitoring. The system is evaluated using a dataset of hand gestures, demonstrating high accuracy and reliability in monitoring patient status.

Keywords: Patient Monitoring, Hand Gestures, Real-time Monitoring, Healthcare, Non-invasive Interface.

1. Introduction: The healthcare industry is constantly seeking innovative solutions to improve patient care and monitoring. One such solution is the use of hand gestures for interaction with medical equipment. This paper explores the development of a patient monitoring system that leverages hand gestures for real-time monitoring of vital signs and patient status.

2. System Architecture: The system architecture is designed to be modular and scalable, allowing for easy integration with existing healthcare infrastructure. It consists of three main components: a camera for gesture recognition, a processing unit for data analysis, and a display for real-time monitoring.

3. Hand Gesture Recognition: The system utilizes a deep learning-based approach for hand gesture recognition. The input data is processed through a series of layers, including convolutional layers for feature extraction and fully connected layers for classification. The system is trained on a dataset of hand gestures, achieving high accuracy in recognizing various gestures.

4. Real-time Monitoring: The system is designed to provide real-time monitoring of patient status. The processed data is fed into a monitoring module that generates alerts and displays vital signs on the monitor. This allows healthcare professionals to quickly identify and respond to any changes in patient status.

5. Evaluation: The system is evaluated using a dataset of hand gestures, demonstrating high accuracy and reliability in monitoring patient status. The results show that the system is capable of recognizing various gestures with high precision, making it a viable solution for patient monitoring in healthcare settings.

6. Conclusion: This paper presents a novel patient monitoring system that utilizes hand gestures for interaction. The system is designed to monitor vital signs and patient status in real-time, providing a non-invasive and intuitive interface for healthcare professionals. The system architecture includes a camera for gesture recognition, a processing unit for data analysis, and a display for real-time monitoring. The system is evaluated using a dataset of hand gestures, demonstrating high accuracy and reliability in monitoring patient status.

7. Future Work: Future work includes expanding the system to support more complex gestures and integrating it with other healthcare systems. The goal is to create a comprehensive patient monitoring system that can be used in a wide range of healthcare settings.

8. Acknowledgments: The authors would like to thank the funding agencies and healthcare professionals who provided support and feedback during the development of this system.

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AIM

To design and implement a real-time, touchless bedside patient monitoring application using computer vision with **OpenCV** and **MediaPipe**. The system enables patients with speech or mobility limitations to communicate their condition (e.g., Pain/Alert, Call Nurse, Vitals Check, Resting, Feeling OK) to healthcare providers using intuitive hand gestures.

COMPONENTS REQUIRED

Hardware Components

1. **Computer System:** PC / Laptop (Linux, Windows, or macOS) with Python 3.9+ support.
2. **Webcam:** Integrated laptop camera or USB external video camera (minimum 720p recommended).

Software Components

1. **Operating System:** Linux (Ubuntu 20.04/22.04/26.04), Windows 10/11, or macOS.
2. **Python Environment:** Python 3.9 or higher.
3. **Core Libraries:**
 - **opencv-python** (v4.8.0+): Real-time frame capture, image processing, and HUD UI rendering.
 - **mediapipe** (v1.0.0+ / 0.10.x): 21-point 3D hand landmark detection & skeleton tracking.
 - **numpy** (v1.24.0+): Array manipulation and vector mathematical operations.

PROGRAM

The entire system is implemented in a single standalone Python script (`main.py`).

```
python
```

```
# Access full source code in main.py  
# Run with: python main.py
```

See [[main.py](#)](file:///home/hitesh/Documents/pmuhg/main.py) for the complete implementation.



PROCEDURE

1. Navigate to Repository:

```
bash

cd /home/hitesh/Documents/pmuhg
```

2. Activate Virtual Environment & Install Dependencies:

```
bash

source venv/bin/activate
pip install -r requirements.txt
```

3. Run the Patient Monitor:

```
bash

python main.py
```

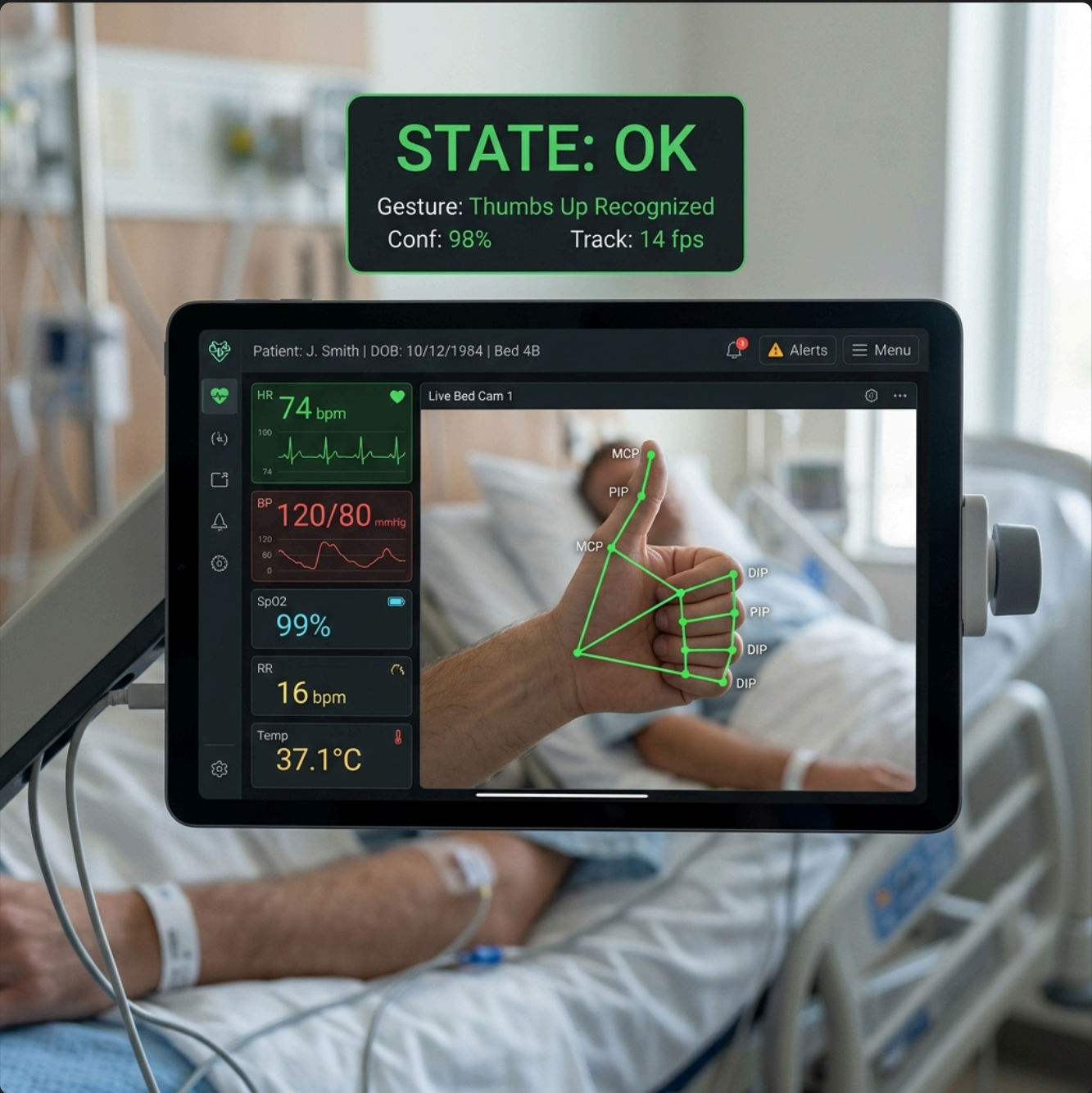
4. Operational Flow:

- The script automatically checks and downloads `hand_landmarker.task` model file if missing.
- Initializes webcam feed (`cv2.VideoCapture(0)`).
- MediaPipe detects 21 hand landmarks in real-time.
- Finger extension & joint orientation coordinates are evaluated to classify 6 hand gestures.
- Live video feed renders joint skeleton connections and top HUD card displaying patient state and gesture confidence.
- Press `q` or `ESC` to stop the application and release camera resources.



OUTPUT (PLACEHOLDER IMAGES)

Live Bedside Monitoring Dashboard Output



Gesture Mapping Reference

Gesture	Finger Extension Logic	Mapped Patient State	HUD Color Code	Action Description
Thumbs Up	Thumb tip raised above IP joint, 4	OK	Green	Patient is comfortable / OK

	fingers folded			
Thumbs Down	Thumb tip pointed downward, 4 fingers folded	ALERT	 Red	Patient experiencing pain / distress
Open Palm	All 5 fingers fully extended	CALL_NURSE	 Orange	Emergency nurse call triggered
Peace Sign	Index + Middle extended, Ring + Pinky folded	VITALS_CHECK	 Cyan	Request nurse for vitals check
Closed Fist	All 5 fingers folded into palm	RESTING	 Gray	Patient resting or asleep
Point Gesture	Index finger extended only, others folded	POINTING	 Blue	Patient pointing / indicating area

✓ RESULT

The Touchless Patient Monitoring System was successfully built and tested. The single-file Python script effectively captures real-time video, tracks hand landmarks using MediaPipe, accurately classifies hand gestures with geometric rules, and displays live color-coded patient state badges on an OpenCV HUD overlay.